

[07 BENG – 2101]

II/IV B.Tech. DEGREE EXAMINATION.

First Semester

Computer Science Engineering

DATA STRUCTURES

(Common with I.T)

(Effective from the admitted batch of 2015–2016)

Time : Three hours

Maximum : 70 marks

First question is compulsory.

Answer any FOUR questions from the remaining questions.

All questions carry equal marks.

Answer all parts of any question at one place.

1.
 - ~~(a)~~ How to simulate recursion in C language?
 - (b) Write the abstract data type for stack.
 - ~~(c)~~ What is doubly linked list?
 - ~~(d)~~ Describe the operations on disjoint sets.
 - (e) Write the algorithmic notation of disjoint sets.

- ~~(f)~~ Draw the decision tree for insertion sort.
- (g) What is transitive and reflexive transitive closures of a graph?
2. (a) Write a function to add two polynomials using abstract data types.
- (b) Write a matrix multiplication function using dynamically allocated arrays.
- ~~3.~~ (a) Write about different types of queues and their operations.
- (b) " $a/b - c + d * e - a * c$ " produce postfix notation for the given infix expression using C language.
4. Write a C program to perform insertion and deletion operations on priority queues using linked list.
5. (a) Write a function that traverses threaded binary tree in post order? Analyze the time space complexity.
- ~~(b)~~ How to split a binary search tree? Explain with a C program.
- ~~(a)~~ What is the role of mid element in binary search? Write a non recursive algorithm for binary search.
- ~~(b)~~ Differentiate sequential search and interpolation search with respect to time and space complexities.

7. (a) What is the role of adjust function in heap sort? Analyze the performance of it.
- (b) What is external ~~sorting~~? Write the C program to sort
25, 5, 77, 1, 61, 11, 59, 15, 48, 19.
8. (a) Write a C program to find out biconnected components of a graph.
- ~~(b)~~ What is minimum cost spanning tree? Write Krushkal algorithm that generates minimum cost spanning tree.
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